



RTL2GDSII WORKSHOP

DAY -5

POWERPLANNING

in

PHYSICAL DESIGN

using

INTEGRATED CIRCUIT COMPILER (ICC2)

Only for chip to startup (C2S) Training

Power Planning in Physical Design

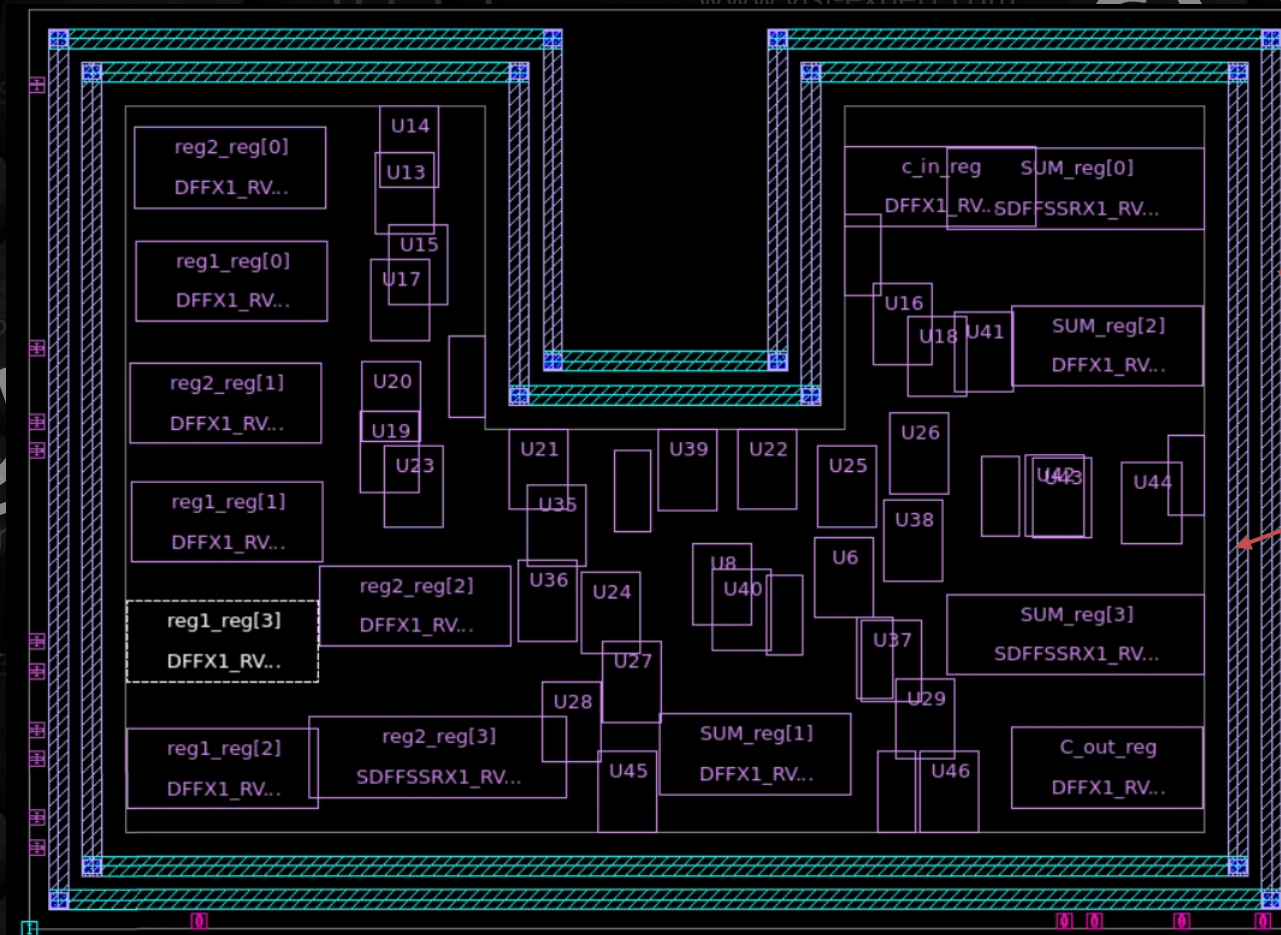
- Powerplanning is an **integral part of Floorplanning stage**.
- Powerplanning is a **Pre-Routing** process.
- **Routing** – Actual Wires (metal) for interconnections.
- **Purpose** : To receive **stable power supply** across chip by creating **Power Distribution/delivery Network (PDN)**
*(Actual Physical (metal) Wire is deployed to construct the **Core Power Ring, Power Mesh and S routes**)*

Steps in Powerplanning

- Creating Power Distribution/delivery Network (PDN)
- Step 1 : Create Core Power Ring

These rings encircle the core area and any macros within the chip.

Power rings carry both the VDD (supply voltage) and VSS (ground) signals.



VDD

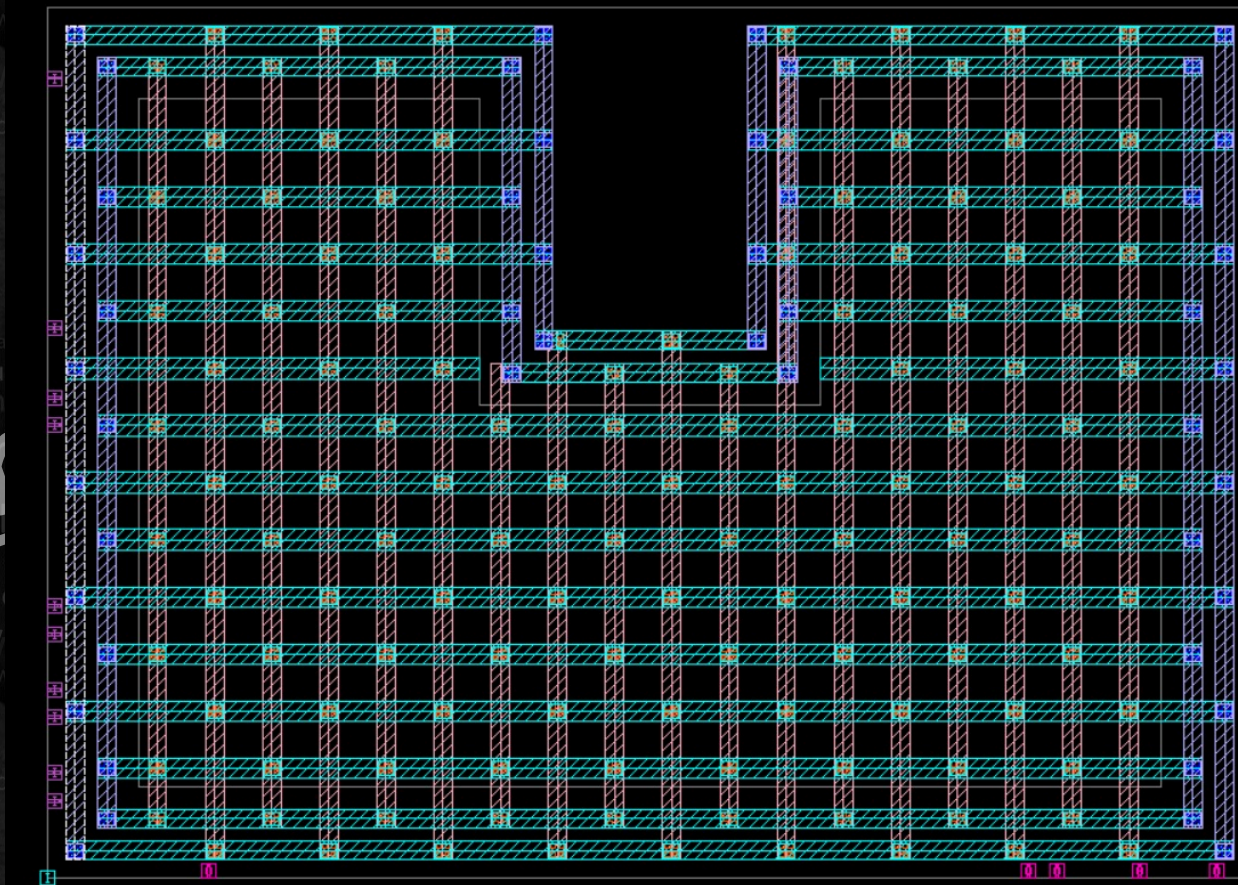
VSS

Only for C

Steps in Powerplanning

- Step 2: Create **Power Mesh/Power Straps/Power Stripes**

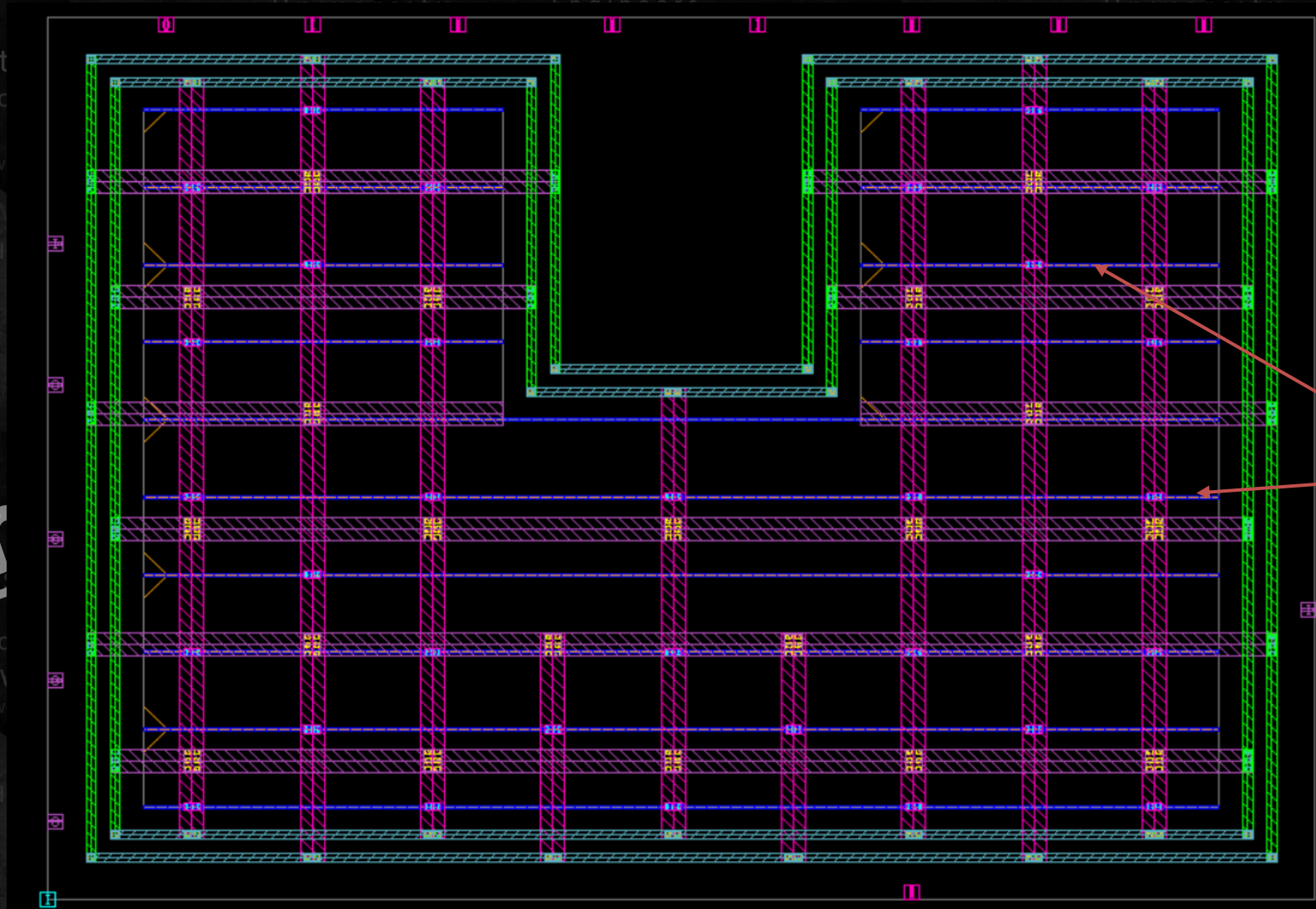
Power straps are Hor./Ver. Structures connecting core ring end to end



Only for

Steps in Powerplanning

- Step 3 : Create Std. Cell route / S routes



Planning

S route

Only for